0580_m24_qp_32_Q6

6 **(a)** Town S is 44 km from town R on a bearing of 117°.

(i) Using a scale of 1 cm represents 8 km, mark the position of town S.

Scale: 1 cm to 8 km

[2]

(ii) Anvi cycles the 44 km from R to S.
She leaves R at 13 15 and cycles at a speed of 12 km/h.

Work out the time she arrives at S.

..... [3]

(b) A tower has a height of 16 metres.
When Jai makes a scale drawing of the tower it has a height of 20 cm.

Work out the scale Jai uses, giving your answer in the form 1 : n.

1 : ..... [2]

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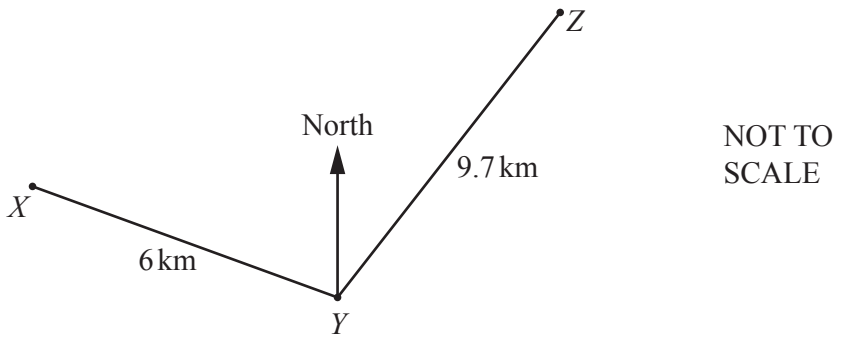
(c) X, Y and Z are three towns.



X is on a bearing of 288° from Y.
Z is on a bearing of 018° from Y.

(i) Show that angle XYZ is 90°.

[2]

(ii) $XY = 6\,\text{km}$ and $YZ = 9.7\,\text{km}$.

Calculate XZ.

$XZ =$ ..... km [2]